Analysis of the 2016, 2020, and 2024 Presidential Elections from an Intersectional Perspective

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Stakeholder Mentor: Prof. Janet Knoedler





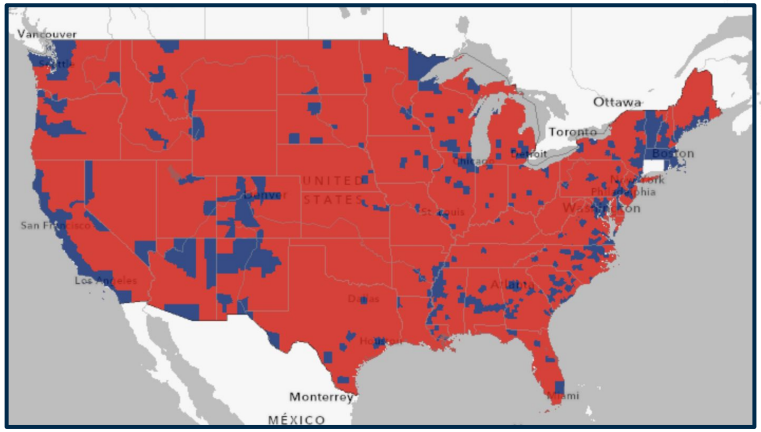
Our Question:

What can we learn from voting data about the 2024 election in comparison to the elections of 2016 and 2020?

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The Project

- **Collected raw data files** on the **2024 Presidential Election** from each state's official website
- **Cleaned the data** to create a **visualization through ArcGIS** of the election data, alongside socioeconomic variables collected by our mentors
- Will be made **publicly available** through Bucknell Digital Commons
- ArcGIS Map



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ECON 204

The ArcGIS map was made for a project in Professor Knoedler's ECON 204 class

Our goal was to use the map to **answer our questions about the previous three elections**

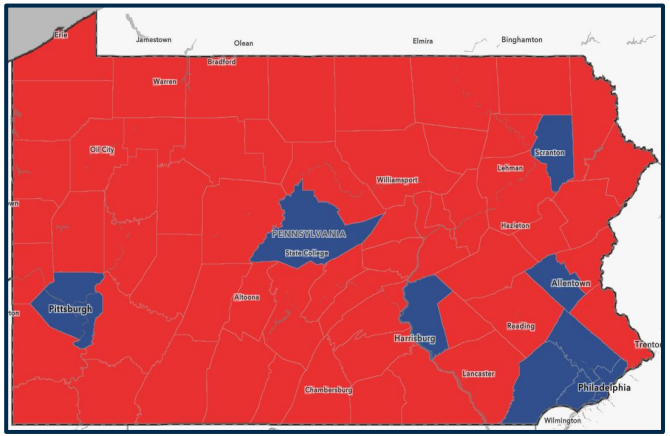
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The Election in Pennsylvania

Had leaned Democratic in previous decades

Recently became a pivotal swing state in Presidential Elections

Donald Trump has won between 48.2% and 50.4% of statewide votes in the last three elections



2024 Presidential Election Results, by County

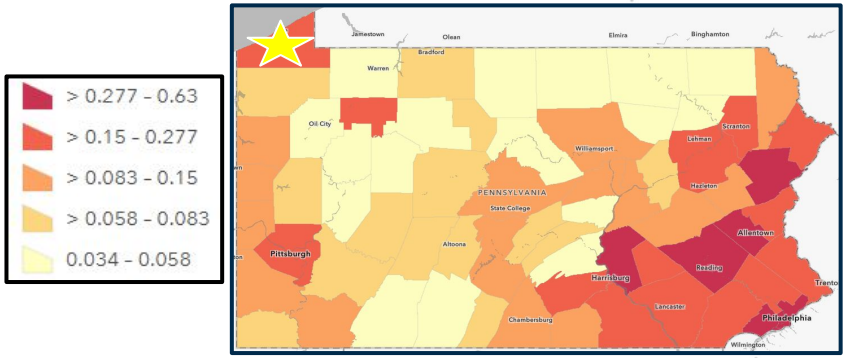
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Percent Non-White Population

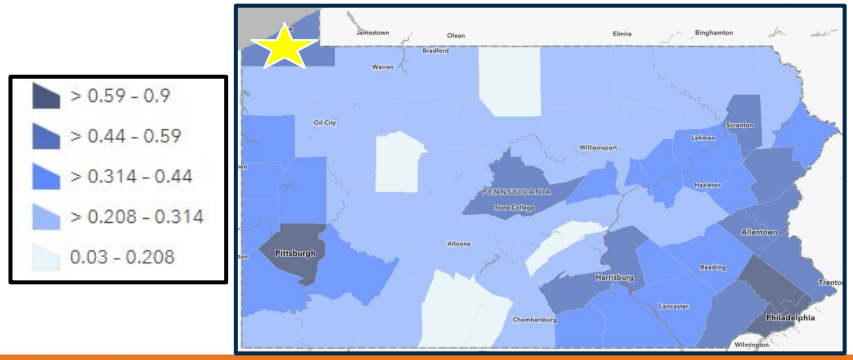
- Kamala Harris **struggled to gain** standard Democratic **support from Non-White populations**
    - i. This could be due to issues of representation (Shell) and not focusing on central issues (Shamim)

- Erie County (starred) swapped from Trump in 2016 to Biden in 2020 to Trump in 2024
    - i. In the last three elections, **the winner never won the majority of the vote**

Percent of Population that is Non-White, by County (2023)



Percent Vote for Harris, by County (2024)

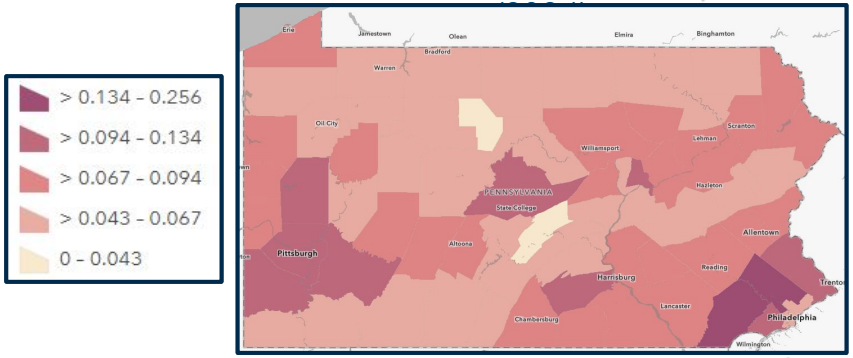


Education of White Women

Strong inverse relationship between percent vote for Trump and percent of women who have obtained a college degree or higher

Generally, the higher educated the women in a county are, the worse Trump did in that county

Percent of White Females with a College Degree or Higher, by County



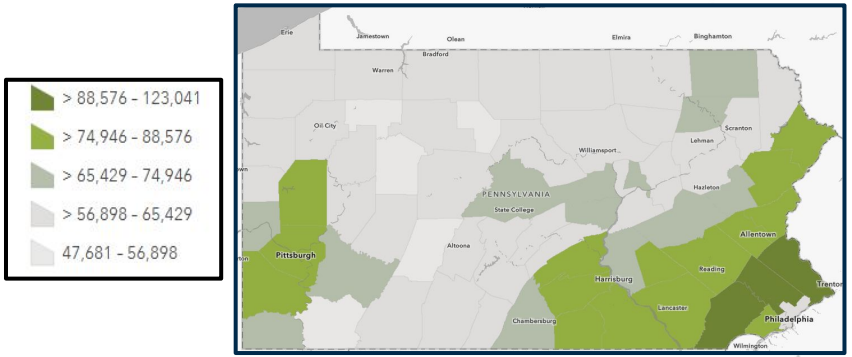
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Median Household Income

Relationship between percent vote for Trump and Median Household Income

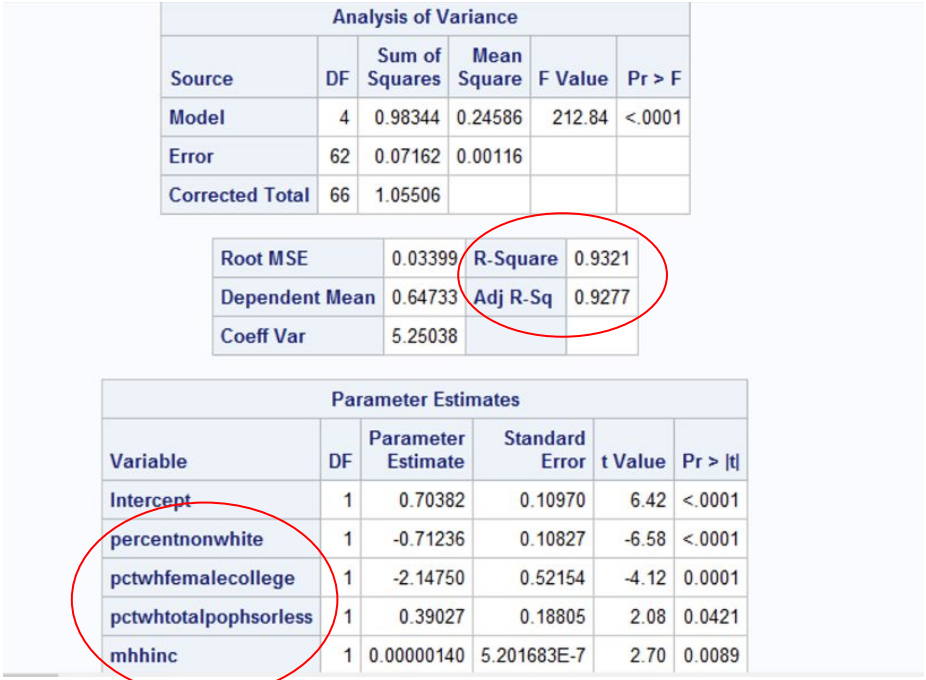
Generally, the higher the median household income in a county, the higher the support for Trump from 2016-2024 in that county

Median Household Income, by County (2023)



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Can we use socioeconomic variables to predict the winner of the election in PA?



Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	0.98344	0.24586	212.84	<.0001
Error	62	0.07162	0.00116		
Corrected Total	66	1.05506			

Root MSE	0.03399	R-Square	0.9321
Dependent Mean	0.64733	Adj R-Sq	0.9277
Coeff Var	5.25038		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > |t|
Intercept	1	0.70382	0.10970	6.42	<.0001
percentnonwhite	1	-0.71236	0.10827	-6.58	<.0001
pctwhfemalecollege	1	-2.14750	0.52154	-4.12	0.0001
pctwhtotalpophsorless	1	0.39027	0.18805	2.08	0.0421
mhhinc	1	0.00000140	5.201683E-7	2.70	0.0089

Maybe?

Decided against doing a logistic regression, due to the small sample size

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Conclusion in Pennsylvania

Republicans are likely to continue their success in Pennsylvania by **leveraging support from counties with lower levels of college attainment**

Democrats will need to **strengthen their appeal to women and minority populations** to increase their competitiveness in Pennsylvania in upcoming elections

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Sources

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Shell, Christopher. “Kamala Harris’s Racial Identity and the Black Electorate.” *Carnegie Endowment for International Peace*, 10 Mar. 2025, carnegieendowment.org/research/2025/03/kamala-harris-racial-identity-and-the-black-electorate?lang=en.

“2024 Presidential Election (Official Returns).” *Commonwealth of Pennsylvania, 5 Nov. 2024, https://www.electionreturns.pa.gov/General/SummaryResults?ElectionID=105&ElectionType=G&IsActive=1*

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